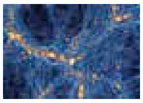
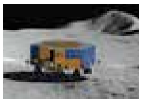


Galaxy simulations
Researchers reports that new galaxy simulations can help reveal the origins of the Milky Way. **http://dgit.in/sep20-69**



Moon mission on SpaceX rocket
A NASA lunar lander mission to the moon's South Pole will launch on a SpaceX rocket in late 2022. **http://dgit.in/sep20-70**

PLATO are focused on finding more exoplanets instead of studying the ones that we already know of. Twinkle will fill the gap till then, and will be dedicated to studying the most promising exoplanets where life as we know it on Earth is likely to exist.

The data received from the existing exoplanet surveys informs us precious little about the planets that they find. From this data, scientists can figure out the mass of the planets, their density and the distance from the planet. Only a few planets have been imaged directly, and even these are a few pixels of data at best. The JWST will be able to directly image more exoplanets, but a dedicated instrument such as Twinkle can tell us so much more by looking into the atmospheres of these worlds and answer many questions about the local conditions on the planet such as the chemical processes that are ongoing in the atmosphere, how it has evolved over time, and the history of the planet. Interactions with other bodies within the system causes changes in the atmosphere, which is also a factor of how close the planet is to the host star. In our system for example, the ravages of the Sun are stripping away the atmosphere of Mars, which does not have an active dynamo that provides a global magnetosphere. Twinkle will be able to detect water vapour, carbon dioxide, carbon monoxide in an atmosphere of an exoplanet, as well as organic molecules such as methane, ethane and ethylene.

Researchers have found many rocky exoplanets in orbit around red dwarf stars, and computer simulations have shown that rocky exoplanets are very likely to form around such stars. While these stars have a narrow goldilocks zone, they are slow burning and can provide light for trillions of years, giving plenty of time for life to take hold. While the Kepler mission has detected many exoplanets in orbit around red dwarf stars, not many rocky exoplanets in orbit around bright Sun-like stars are known. The TESS mission is expected to find more such exoplanets, and by the time it does, Twinkle will be ready to characterise them. Twinkle is the first mission that will be launched with the specific task of studying the atmospheres of exoplanets.

The satellite is an international collaboration, and is being assembled in the UK. A platform by Surrey Satellite Technology is being used, along with instrumentation by the University College of London. Researchers from Ohio State University and Vanderbilt University in the US are putting together a list of candidates for the Twinkle mission to study. Ji Wang, assistant professor of astronomy at Ohio State University and one of the people identifying the targets for study by Twinkle says, "Twinkle is nimble, so we will be able to observe many candidates with multiple visits to the same target. We also have a very competitive team here at Ohio State that can leverage NASA's James Webb Space Telescope mission to focus on a handful of the most promising targets identified for biosignature detection. Finding biosignatures on exoplanets represents one of our best chances of finding signs of life elsewhere."

Twinkle will be deployed at a Low Earth Sun Synchronous orbit, at an altitude between 600 to 700 kilometers. The satellite will orbit the Earth in such a way, that it will perpetually be in either dawn or dusk, which maximises the time available for conducting observations. The planned orbit period is between 90 to 100 minutes, which is how long it will take for each orbit around the planet. The light for the observations will be captured by a Rutherford Appleton Laboratory Camera-4 (RALCAM4), equipped with a steerable mirror to compensate for tiny movements during the observation. From here the light will be filtered, and split between two spectrometers. One of these is called Exoplanet Light Visible Spectrometer (ELVIS), and is based on the spectrometer used by the Trace Gas Orbiter (TGO) on Mars. The other one is an infrared spectrometer for studying super Earths and hot Jupiters, and is designed on similar lines as the Mid Infrared Instrument on the JWST. Twinkle is slated for a 2021 launch. d

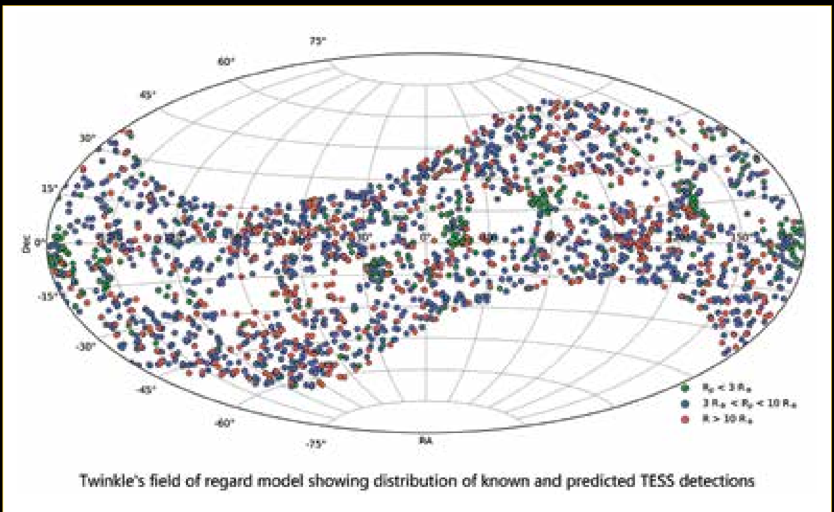


Image: Twinkle Mission

Twinkle's field of view marked with predicted candidates expected to be discovered by TESS